Data Changes Everything

Name _____

Work with a partner. Compare answers when you finish each problem.

Word bank: 14 million 15% 26% 167 20,000 49,000 chips data GPUs label neural network video games

You can use a word more than once.

1. How did you learn?

I learned to recognize _____ from **rules** / **examples** (circle one).

2. ImageNet.

14 million
pictures

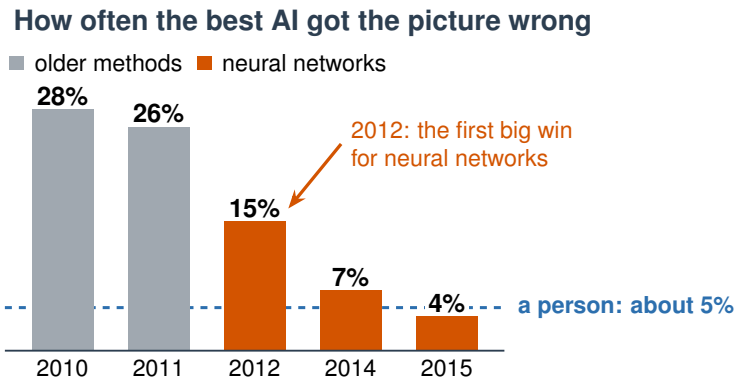
20,000+
kinds of things

49,000
people in 167 countries labeled them

ImageNet had about _____ pictures of more than _____ kinds of things.

_____ people in _____ countries looked at the pictures and gave each one a _____.

3. The big win in 2012.



In 2012, a _____ won the contest. Its mistakes dropped from about _____ to _____.

It was trained on _____: computer chips made for _____.

The new recipe for AI: 1. lots of _____ 2. _____ 3. fast _____

4. Be the labelers. Do the sandwich round. Label alone first. Then compare with a partner.

Round	**Pictures I said YES to** (numbers)	**Pictures we disagreed on**
sandwich		
chair (Day 2)		
car (Day 2)		

Class vote. The hardest picture was _____. Yes: _____ No: _____

An AI learns what the _____ decided. It does not know what a sandwich *really* is.

When I click every picture with a traffic light in a CAPTCHA, I am a _____ for AI.

5. Talk about it. 49,000 people labeled ImageNet. Most people do not know their names.

The people who labeled the data **should** / **should not** get credit, because _____

_____

Go Further (Day 2). Do the chair and car rounds in the table above.

My rule for "sandwich": It is a sandwich if _____

✂ *Cut out the cards. Add them to the class timeline.*

2009

Fei-Fei Li's team shares **ImageNet**: 14 million labeled pictures.

Draw a picture here.

2012

A **neural network** trained on video game chips wins the ImageNet contest.

Draw a picture here.

2016

AlphaGo beats Lee Sedol at Go, 4 games to 1.

Draw a picture here.